Aprendizaje No Supervisado y Texto como Datos

Big Data y Machine Learning para Economía Aplicada

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Agenda

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Motivation

$f(X\Omega) \longrightarrow \hat{y} \qquad x$

- One way to think about almost everything we do is as dimension reduction.
- We are trying to learn from high-dimensional X some low-dimensional summaries that contain the information necessary to make good decisions.
- We have a high-dimensional X, and you try to model it as having been generated from a small number of components/factors.
- We are attempting to simplify X for its own sake.

$|\widehat{\text{precio}} - \text{precio}|$

~~Superar zale.~~

Motivation

$y =$ precio $|y - \hat{y}|$

$(y - \hat{y})^2$

- Unsupervised learning is often much more challenging.
- The exercise tends to be more subjective, and there is no simple goal for the analysis, such as prediction of a response.
- Unsupervised learning is often performed as part of an exploratory data analysis.
- Furthermore, it can be hard to assess the results obtained from unsupervised learning methods, since there is no universally accepted mechanism for performing cross- validation or validating results on an independent data set.
- There is no way to check our work because we don't know the true answer: the problem is unsupervised.

Es útil ? no

Agenda

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Principal Component Analysis

$\boxed{X_{n\times p}}$

reducir la dimensión
p es grande tratar de reducirla

$n << p \rightarrow$ OLS

- PCA is an unsupervised learning technique that allows to
  - reduce the dimensionality of data sets,
  - while preserving as much "variability" as possible.

informativa.

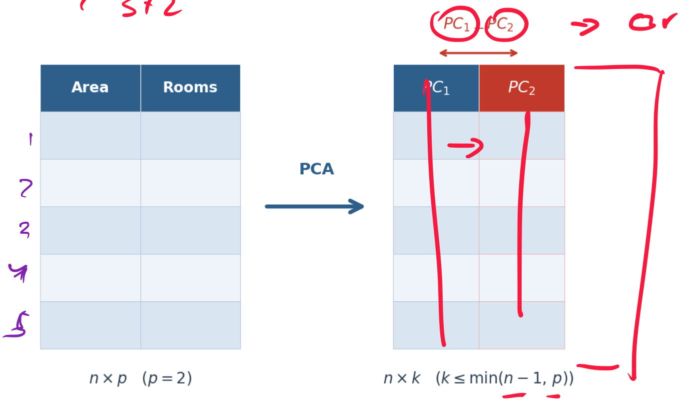
- It is an unsupervised approach, it involves only a set of variables/features $X_1, X_2, \ldots, X_p$, and no associated response Y

Principal Component Analysis

What it does

$X_{5 \times 2}$ → Componentes Z

$PC_1 \perp PC_2$ → ortogonales

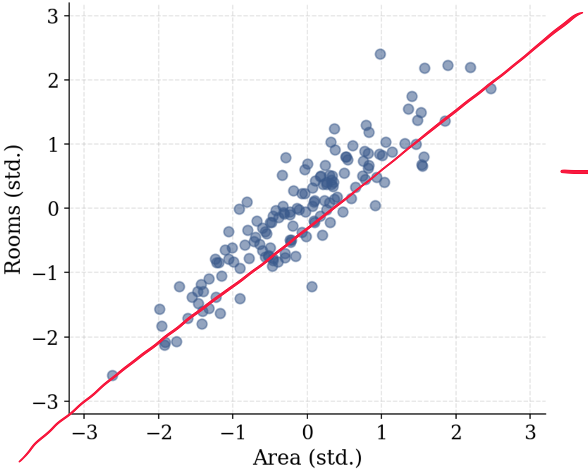


Principal Component Analysis

- PCA reprojects the data from the original p-dimensional space into a new coordinate system (a new space).
- The new axes, the principal components,are ordered by the amount of information (variance) they capture.
- Once reprojected, the analyst chooses to retain only the first k components, this is where dimensionality reduction happens.

Principal Component Analysis

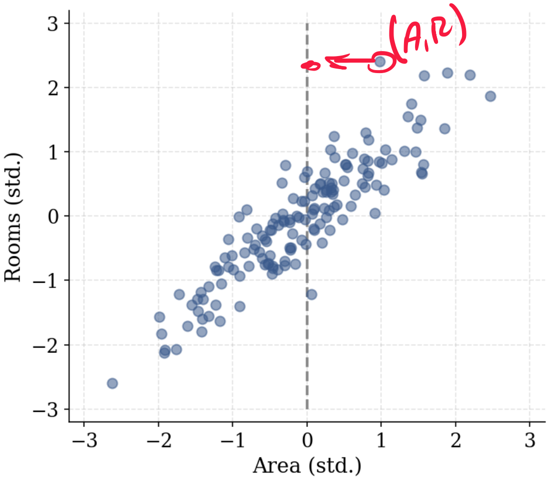
$$\frac{x - \mu}{s}$$



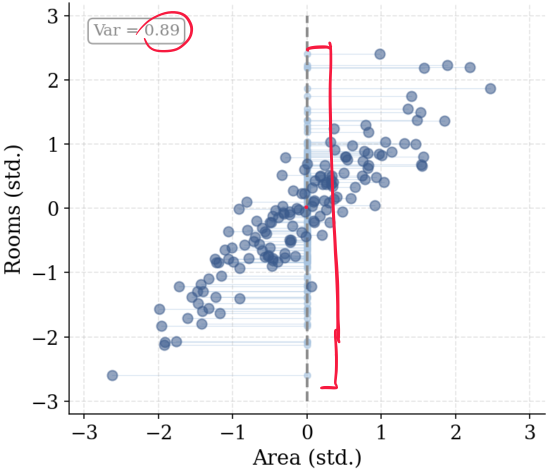
→ Reproyector.

PC1

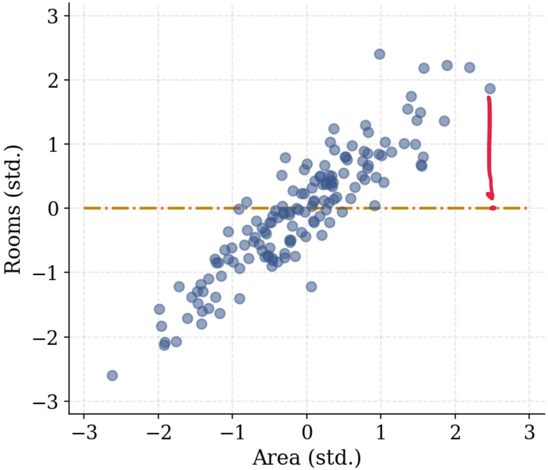
Principal Component Analysis



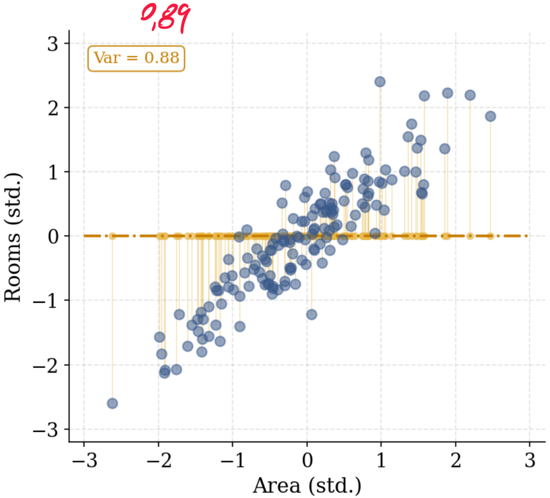
Principal Component Analysis



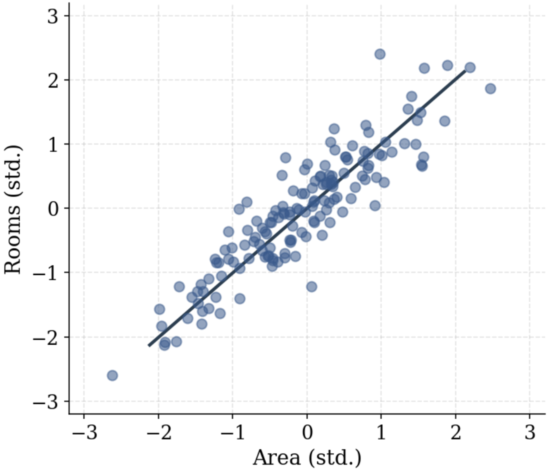
Principal Component Analysis



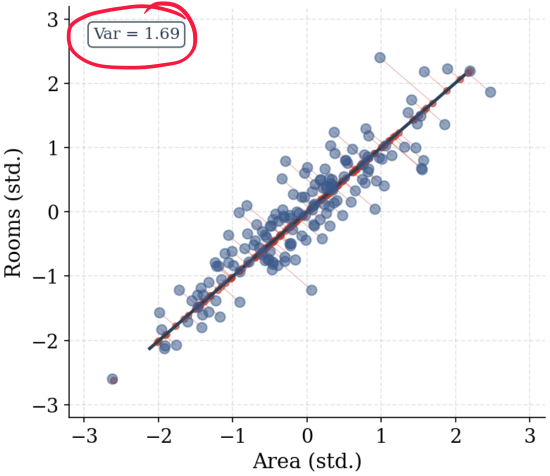
Principal Component Analysis



Principal Component Analysis



Principal Component Analysis

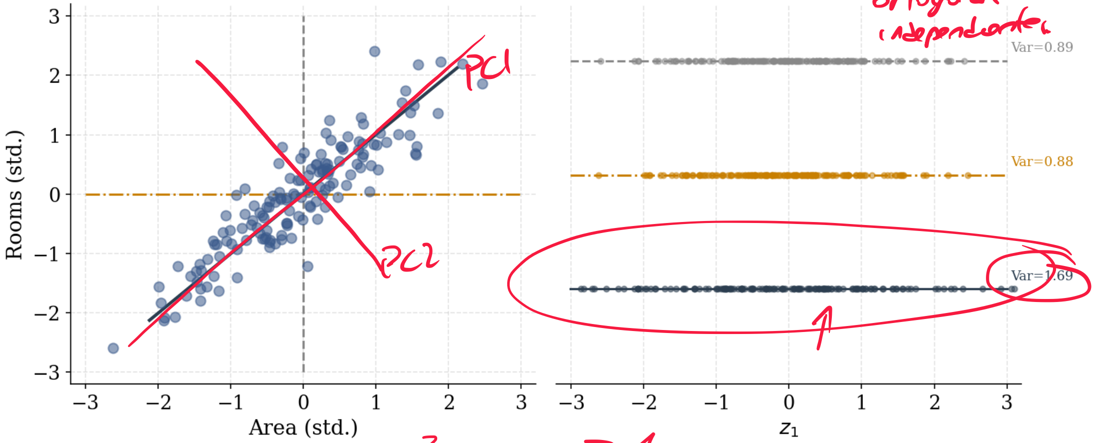


Principal Component Analysis

$X_{n\times 2} \rightarrow$ PC1 $\perp$ PC2

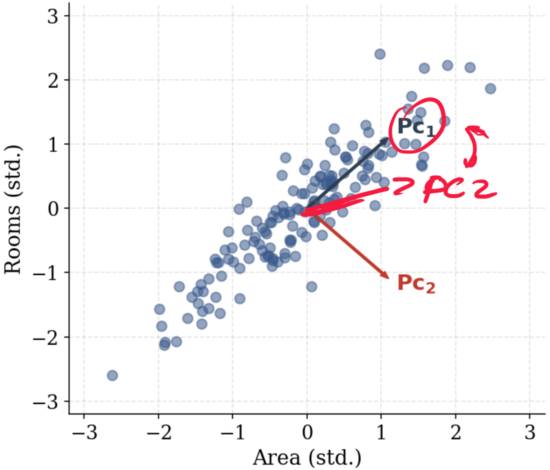
Room = $\alpha + \beta$ Area

ortogonal
independiente



$2 \rightarrow 1$

Principal Component Analysis



Principal Component Analysis

- The first principal component of a set of features $X_1, X_2, \ldots, X_p$ is the normalized linear combination of the features

$$PC_1 = \delta_{11} X_1 + \delta_{21} X_2 + \cdots + \delta_{p1} X_p \tag{1}$$

- The δ coefficients are called loadings or rotations

weights

- By normalized we mean that $\boxed{\sum_{j=1}^{p} \delta_{j1}^2 = 1}$

$\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

- In our example:

$$PC_1 = \overset{\frac{1}{2}}{\delta_{11}} Area + \overset{\frac{1}{2}}{\delta_{21}} Rooms \tag{2}$$

Motivation

$Var(aX) = a^2 Var(X)$

- Given a $n \times p$ data set X, how do we compute the first principal component?

$PC1 = \max Var(X\delta_1)$

$$PC_1 = \boxed{X\delta_1} \tag{3}$$
$$= \delta_{11}X_1 + \delta_{21}X_2 + \cdots + \delta_{p1}X_p \tag{4}$$

$\delta_1' \underbrace{Var(X)}_{S} \cdot \delta_1$

- The idea is to preserve the most information possible
- In other words, we are going to try to generate an index that reproduces (the best it can) the information (variability) of the original variables
- How we do that? $\to$ maximize the variance

λ mas grande

autovector

Principal Component Analysis

- PCA solves step by step

S = Varianza de X (estandarizada)

$$
\begin{aligned}
&\max_{\delta}\ \delta_1' S \delta_1 \\
&\text{subject to} \\
&\delta_1'\delta_1 = 1
\end{aligned}
\tag{5}
$$

$= \sum \delta_{ij}^2 = 1$

- Let us call the solution to this problem δ_1^*.
- $PC_1^* = X\delta_1^*$ is the 'best' linear combination of X.
- Intuition: X has p columns and $PC_1^* = X\delta_1^*$ has only one. The factor built with the first principal component is the best way to represent the p variables of X using a single variable.

Principal Component Analysis

- Solution to the problem of the first principal component

Principal Component Analysis

- Solution to the problem of the first principal component
- Let's set the lagrangian

$$\mathcal{L} = \delta_1' S \delta_1 + \lambda_1 (1 - \delta_1' \delta_1) \tag{6}$$

- Rearranging

$$S\delta_1 = \lambda_1 \delta_1 \tag{7}$$

- At the optimum, δ is the eigenvector corresponding to the eigenvalue λ of S.
- Premultiplying by δ_1' and remembering that $\delta_1'\delta_1 = 1$:

Principal Component Analysis

$\delta_1' S \delta_1 = \lambda_1$

$X \rightarrow$ estand $\frac{X-\mu}{desest}$

$$\delta_1 S \delta_1 = \lambda_1 \tag{8}$$

$Var(X) = S \rightarrow$ autovalores

- In order to maximize $\delta' S \delta$ we must choose λ equal to the maximum eigenvalue of S and δ is the corresponding eigenvector.
- The problem of finding the best linear combination that reproduces the variability of X is finding the biggest eigenvalue of S and it's corresponding eigenvector

Principal Component Analysis

x_1
x_2

- Having found PC_1^*, we ask: is there a second component?
- Yes! but we impose $\delta_2'\delta_1 = 0$:

$PC_2 = \delta_{12} x_1 + \delta_{22} x_2 + \dots + \delta_{p2} x_p$

$$\max_{\delta_2} \delta_2' S \delta_2 \qquad \text{s.t.} \quad \delta_2'\delta_2 = 1, \quad \boxed{\delta_2'\delta_1 = 0} \tag{9}$$

→ segundo autovalor + grande

- **Why orthogonality?** If δ_2 were not orthogonal to δ_1, PC_2 would partially re-explain the same variation already captured by PC_1. The constraint $\delta_2'\delta_1 = 0$ guarantees that PC_2 captures *only new* variation. This implies $\text{Cov}(PC_1^*, PC_2^*) = 0$.
- $PC_2^* = X\delta_2^*$ is the best linear combination of X subject to being uncorrelated with PC_1^*.
- Recursively, this logic yields up to $\min(n-1, p)$ components, each orthogonal to all previous ones.

Selection of factors

Var % % St 2 St 2

- Although a matrix X of dimension $n \times p$ generally has $min(n-1, p)$ different principal components.

Corr.

- In practice, we are generally not interested in all the components, but rather stay with the first ones that allow us to visualize or interpret data.
- Indeed, we would like to keep the minimum number that allows us a good understanding of the data.
- The natural question that arises here is whether there is an established way to determine the number of principal components to use.
- Unfortunately, there is no accepted objective way in the literature to answer it.

Selection of factors

X ; Z $\boxed{1 \mid 1}$ $\to$ $\lambda_1 \quad \lambda_2$

- However, there are three simple approaches that can guide you in deciding the number of relevant major components.
  - Proportion of variance explained. ejemplo.
  - Kaiser criterion.
  - Visual examination of screeplot

$$E(z) = 0$$

$$z = \frac{x - \mu}{\text{desvio estandar}} \to Var(z) = 1$$

Selection of Factors

Kaiser Criterion

- Let the columns of X be standardized, so each variable has unit variance.
- We can show that $\sum_{j=1}^{p} \lambda_j = \sum_{j=1}^{p} V(PC_j)$, and each standardized variable contributes one unit of variance:

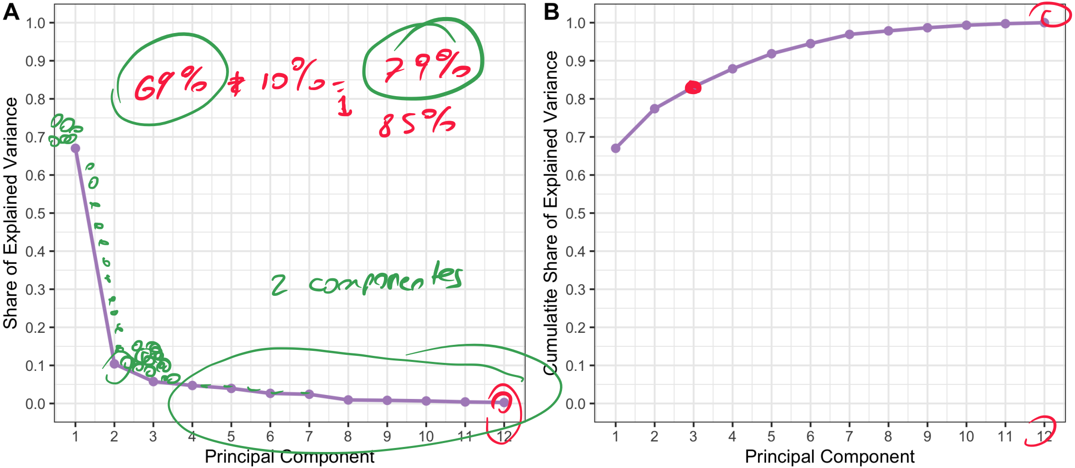
$$\sum_{j=1}^{p} \lambda_j = p \tag{10}$$

- Each original variable accounts for exactly one unit of variance.
- **Kaiser criterion:** retain PC_j if $\lambda_j > 1$. → Aquellos q explican + varianza q el promedio
- Intuition: a component with $\lambda_j \leq 1$ explains *less* variance than a single original variable. It is not worth keeping a summary that does worse than what we started with.

Selection of factors

$X \to 2 \to PC$

Screeplot



Agenda

PCA as a Factor Model

- Some researchers interpret the principal components as *latent factors*: unobserved variables that drive the co-movement in the data.
- The idea is simple: we observe many variables x_i, but behind them there may be a small number of underlying forces explaining why they move together.
- Formally, we write this as:

falta mayuscula.

\delta

$$x_i = \Delta f_i + u_i \tag{11}$$

\Delta

where f_i is $k \times 1$ vector of latent factors and Δ is the $p \times k$ loading matrix, exactly the first k eigenvectors we already found.

PCA as a Factor Model

Test Scores

- $x_i \in \mathbb{R}^p$: the p exam scores of student i
- f_i: the latent ability of student i, never observed directly
- δ_j: how ability loads onto exam j
  - Some exams are highly related to ability $\rightarrow$ large $|\delta_j|$
  - Some exams are weakly related $\rightarrow \delta_j \approx 0$
- With $k > 1$ factors, ability is multidimensional. For example, literary and math ability are different latent forces that load differently onto language vs. math exams.
- In labor economics, the distinction between cognitive and non-cognitive ability has been key to explain wage patterns: some jobs require one, some the other, and some both (e.g., a surgeon needs both).
- We will see this concretely in the notebook.

Factor Interpretation: Examples

Agenda

1. Motivation
2. Principal Component Analysis
   - What are PCAs?
   - PCA as a Factor Model
3. Text as Data
   - Bag of Words
   - Modelado de Tópicos
     - PCA y Modelado de Tópicos
     - LDA

Text as Data: The Big Picture

- We generate vast quantities of raw unstructured text.
- As the costs of storage drop and as more conversations and records move to digital platforms, we accumulate massive corpora that track communications:
  - customer conversations,
  - product descriptions or reviews,
  - news,
  - comments, blogs, tweets, etc...
- The information in text is a rich complement to the more structured variables contained in a traditional transaction or customer database.
- Social scientists have also woken up to the potential of such data and recent years have seen an explosion in studies that make use of text as data.

Text as Data: The Big Picture

- **Text is a vast source of data for research, business,etc**
- It comes connected to interesting “author” variables
  - What you buy, what you watch, your reviews
  - Group membership, who you represent, who you email
  - Market behavior, macro trends, the weather

Text as Data: Motivation

THE JOURNAL OF FINANCE • VOL. LXII, NO. 3 • JUNE 2007

Giving Content to Investor Sentiment: The Role of Media in the Stock Market

PAUL C. TETLOCK*

ABSTRACT

I quantitatively measure the interactions between the media and the stock market using daily content from a popular *Wall Street Journal* column. I find that high media pessimism predicts downward pressure on market prices followed by a reversion to fundamentals, and unusually high or low pessimism predicts high market trading volume. These and similar results are consistent with theoretical models of noise and liquidity traders, and are inconsistent with theories of media content as a proxy for new information about fundamental asset values, as a proxy for market volatility, or as a sideshow with no relationship to asset markets.

Agenda

Text as Data

X → números

+1 a favor
0 → ausente
−1 en contra

- To analyze text, we need to transform it into data that can be input to numeric regression and factorization algorithms.
  - Bag of Words (BoW) and DTMs
  - Word Embeddings

~~LSM~~

$y \sim X$ Descripción

caract

Text as Data

- A passage in '*As You Like It*' from Shakepeare:

  All the world's a stage,
  and all the men and women merely players:
  they have their exits and their entrances;
  and one man in his time plays many parts...

Text as Data

- A passage in '*As You Like It*' from Shakepeare:

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- What the econometrian sees tokens:

world	stage	men	women	play	exit	entrance	time
1	1	2	1	2	1	1	1

From Text to Tokens

- Tokenization = splitting text into units
- Most often: words
- Example:
  "The economy is growing fast" $\rightarrow$ [the, economy, is, growing, fast]
- Tokens are the building blocks for Bag-of-Words

Why Word Counts?

- Texts are turned into signals, words to numbers
- **Distributional Hypothesis** (Harris, 1954): words in similar contexts have similar meanings
- Documents with similar word distributions likely discuss similar topics
- Counts give a first approximation of meaning

Counts as Approximation of Meaning

- Simplify: ignore order and grammar, keep word frequencies
- Frequent words reflect themes and emphasis
- Analogy to economics: start parsimonious, add frictions later

Bag-of-Words (BoW) Idea

- Once text has been tokenized
- Ignore grammar, word order, syntax, and context
- Represent each document as a vector of counts:

Portin

mesl

$$d = (c_1, c_2, \ldots, c_V)$$

- What is lost: structure and nuance
- What is kept: themes and emphasis
- Philosophy: sacrifice structure, keep signal

From Document to Corpus

descripcion de la propieda!

- **Document**: a single text (article, tweet, speech, passage)
- Represented as a vector of word counts after tokenization

$$d = (c_1, c_2, \ldots, c_V)$$

- **Corpus**: a collection of documents
- Examples:
  - All Shakespeare plays
  - A set of news articles
  - Student essays in this class

descripcion de todos los prop.

Document-Term Matrix (DTM)

- Apply Bag-of-Words to every document in a corpus
- Stack results into a matrix:

DTM

$$\cancel{X} = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1V} \\ c_{21} & c_{22} & \cdots & c_{2V} \\ \vdots & \vdots & \ddots & \vdots \\ c_{D1} & c_{D2} & \cdots & c_{DV} \end{bmatrix}$$

- Rows = documents (D)
- Columns = terms (V words in the vocabulary)
- Entries c_{ij} = count of word j in document i

Text as Data: BOW

- *Juan tiene una vaca. La vaca es buena.* d1
- *Pedro tiene un caballo. El caballo es indomable.* d2
- *La Economía va a tener una desaceleración.* d3

DTM (rows = docs, cols = vocab):

	Juan	Pedro	La	Economía	tiene	va	a	tener	una	un	vaca	caballo	desaceleración	es	buena	indomable
d_1	1	0	0	0	1	0	0	0	1	0	1	0	0	1	1	0
d_2	0	1	0	0	1	0	0	0	0	1	0	1	0	1	0	1
d_3	0	0	1	1	0	1	1	1	1	0	0	0	1	0	0	0

3x16

Text as Data: Cleaning and Tokenization

- Not all words are useful in the DTM
- **Cleaning:** remove elements that do not add meaning to content or structure
- There is no single recipe, depends on purpose and source

Cleaning: Common Steps

ojo GREP

- Convert to lowercase, drop numbers and punctuation
  - But context matters: don't drop :-) from tweets!
- Remove **stopwords** (if, and, but, the, they, . . .)
  - Caution: one person's stopword may be another's key term
- Remove rare words (very low frequency across docs)

Stemming vs Lemmatization

run , runs → ru

S

- Goal: reduce words to a base form
- **Stemming:** crude rules, chop endings → basa en reglas → rápido
- **Lemmatization:** uses vocabulary and grammar → es muy lento (para ustedes no se preocupen)
- Example: *organize*, *organizes*, *organizing* → $\boxed{\textit{organize}}$

boys → boy

girls → girl

→ DNP (lematizador colombiano)

↳ Python

Stemming vs Lemmatization: Examples

- am, are, is $\rightarrow$ be
- car, cars, car's, cars' $\rightarrow$ car
- the boy's cars are different colors
  - Stemming: boy car ~~be~~ differ color
  - Lemmatization: boy car be different color

Stemming in Practice

- **Stemming** = rule-based process that chops word endings
- **English:**
  - Porter Stemmer (classic, simple rules)
  - Snowball Stemmer (successor, more flexible)
  - Lancaster Stemmer (more aggressive, often too short)
- **Spanish:**
  - Snowball Stemmer for Spanish (handles gender and plural forms)
  - Example: *niños, niñas, niño, niña* → *niñ*
- Pros: fast, language-specific implementations
- Cons: crude, may create non-words (e.g., *relational* → *relat*)

Lemmatization in Practice

- **Lemmatization** = reduce to dictionary form (lemma) using vocabulary + grammar
- Uses part-of-speech tagging to disambiguate
- **English:**
  - WordNet Lemmatizer (NLTK)
  - spaCy's lemmatizer (state-of-the-art, context-aware)
- **Spanish:**
  - spaCy Spanish models (es_core_news_sm, md, lg)
  - Freeling (NLP library for Spanish/Catalan)
- Pros: returns valid words, handles meaning and context
- Cons: slower, requires linguistic resources (POS tagging, dictionaries)

Stemming vs Lemmatization: Comparison

Prueben con Ambos

whatever works works!!

	Stemming	**Lemmatization**
Method	Rule-based (chop endings)	Dictionary + grammar
Output	Often non-words (*relat*)	Valid words (*relation*)
Speed	Fast, lightweight	Slower, needs resources
Accuracy	Crude, may conflate terms	Context-aware, more precise
English tools	Porter, Snowball, Lancaster	WordNet, spaCy
Spanish tools	Snowball (Spanish rules)	spaCy (es_core_news), Freeling
When should I use it?	Large-scale, noisy corpora	When interpretability matters

Trade-off: speed vs. accuracy

Bag-of-Words: Strengths and Limits

- **Strengths**
  - Simple, fast, often a strong baseline
  - Captures themes through word choice and frequency
- **Limitations**
  - Ignores context (*bank* = river or money)
  - Ignores order (“man bites dog” = “dog bites man”)
  - Very high-dimensional and sparse

N-grams: Adding Local Order

- An **n-gram** is a sequence of n consecutive words
- Examples:
  - Unigrams: economy, growth
  - Bigrams: central bank, machine learning
  - Trigrams: New York City, stock market crash
- Capture short-range context and common phrases

New York City

N-grams: Trade-offs

clase machine learning

clase. machine machine. learning

- Vocabulary size grows quickly with n (V^n possible n-grams) Check 1.1e
- Many n-grams are rare $\rightarrow$ sparsity increases
- Only capture local context, not long-distance dependencies
- Still a useful bridge between Bag-of-Words and advanced models

Putting It Together

- **Cleaning:** remove noise and normalize
- **Bag-of-Words:** simplest representation, counts

DTM

- **N-grams:** enrich with short sequences
- **Next:** reduce dimensionality, uncover structure (topics, embeddings)

Example

Gentzkow and Shapiro: What drives media slant? Evidence from U.S. daily newspapers (*Econometrica*, 2010)

Econometrica, Vol. 78, No. 1 (January, 2010), 35–71

WHAT DRIVES MEDIA SLANT? EVIDENCE FROM U.S. DAILY NEWSPAPERS

BY MATTHEW GENTZKOW AND JESSE M. SHAPIRO[1]

We construct a new index of media slant that measures the similarity of a news outlet's language to that of a congressional Republican or Democrat. We estimate a model of newspaper demand that incorporates slant explicitly, estimate the slant that would be chosen if newspapers independently maximized their own profits, and compare these profit-maximizing points with firms' actual choices. We find that readers have an economically significant preference for like-minded news. Firms respond strongly to consumer preferences, which account for roughly 20 percent of the variation in measured slant in our sample. By contrast, the identity of a newspaper's owner explains far less of the variation in slant.

KEYWORDS: Bias, text categorization, media ownership.

Example

Gentzkow and Shapiro: What drives media slant? Evidence from U.S. daily newspapers (*Econometrica*, 2010)

- Build an economic model for newspaper demand that incorporates political partisanship (Republican vs Democrat)
  - What would be independent profit-maximizing "slant"?
  - Compare this to slant estimated from newspaper text.
- Use data from Congress to isolate the phrases
- Compare phrase frequencies in the newspaper with phrase frequencies in the 2005 Congressional Record to identify whether the newspaper's language is more similar to that of a congressional Republican or a congressional Democrat

Republican	Democratic
death tax	estate tax
tax relief	tax break
personal account	private account
war on terror	war in Iraq

Text as Data: Example Gentzkow and Shapiro

Agenda

El problema: descubrir de qué se habla

Supongamos que tenemos un conjunto de documentos: artículos de prensa, reseñas, o ensayos. Cada documento contiene miles de palabras, y lo que nos gustaría saber es algo tan simple como:

> *¿Cuáles son los temas principales de este conjunto de textos?*

Queremos, por ejemplo, descubrir que unos textos hablan de política, otros de deportes, y otros de economía, **sin leerlos uno a uno**.

Esto es lo que llamamos **modelado de tópicos** (topic modeling): encontrar patrones de palabras que tienden a aparecer juntas y que nos permiten describir los textos en pocas dimensiones conceptuales.

Nuestra representación: texto como datos

Sabemos que cada documento puede representarse como un vector de palabras. Si construimos una **matriz documento término (DTM)**, tenemos:

- Cada fila = un documento.
- Cada columna = una palabra.
- Cada celda = cuántas veces aparece esa palabra en ese documento (o su peso TF-IDF).

Esta matriz es **muy grande y muy poco densa** (casi todo son ceros), pero contiene toda la información sobre qué palabras co-ocurren en qué documentos.

¿Qué sería un “tópico”?

En esta representación, un **tópico** puede pensarse como un patrón recurrente de uso de palabras:

- El tema *deportes* podría tener alta frecuencia en palabras como *gol*, *equipo*, *partido*, *estadio*.
- El tema *economía* podría estar asociado a *precio*, *inflación*, *mercado*, *inversión*.

Cada documento entonces puede verse como una **combinación de tópicos**: algunos hablan mucho de uno, otros mezclan varios.

Cómo buscar esos patrones: una mirada algebraica

Una forma de pensar este problema es la siguiente:

- Tenemos una matriz enorme X (documentos $\times$ palabras).
- Queremos encontrar **combinaciones de palabras** que expliquen las **mayores diferencias entre documentos**.
- Dicho de otro modo, buscamos direcciones en el espacio de palabras donde los documentos se separen más.

Y eso, en estadística y econometría, tiene un nombre conocido: **maximizar la varianza**.

PCA y Modelado de Tópicos

- Imaginemos los documentos como puntos en un espacio donde cada eje es una palabra. Si dos palabras suelen aparecer juntas (por ejemplo *precio* y *mercado*), los puntos estarán alineados en una dirección particular.
- PCA busca precisamente eso:

  La dirección en la que los documentos difieren más, es decir, donde la ***varianza*** *es máxima.*
- Esa dirección, una combinación lineal de palabras, puede interpretarse como un **tema** o **tópico latente**.

PCA y Modelado de Tópicos

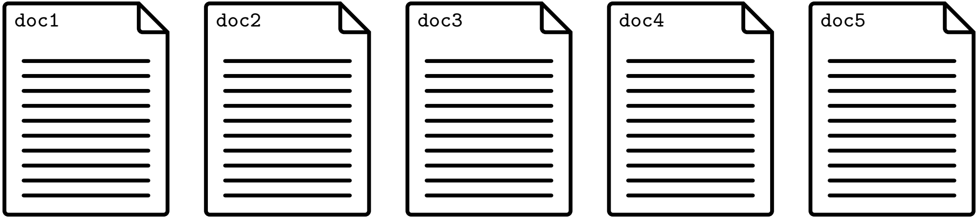
- La utilidad de la reducción de dimensionalidad puede no ser obvia en pocas dimensiones, pero se vuelve mucho más clara cuando tenemos datos altamente dimensionales.
- Por ejemplo, consideremos 500 artículos de prensa de los principales medios colombianos (El Tiempo, El Espectador, Semana) publicados durante 2024, donde cada artículo se representa mediante un vocabulario de 2,000 palabras únicas más frecuentes (excluyendo stopwords). Aquí, n serían los 500 artículos y k las 2,000 palabras del vocabulario
- Necesitamos entonces un método que nos permita encontrar una representación de baja dimensionalidad que contenga la mayor información posible.
- Intuitivamente, el PCA plantea que cada observación se encuentra en un espacio k-dimensional, pero no todas estas dimensiones son igualmente informativas.
- Por lo tanto, podemos usar PCA para representar, o mejor dicho, proyectar, los datos originales a un espacio de menor dimensión de forma tal de retener la mayor cantidad de información posible.

LDA como Modelos de Tópicos

- El enfoque de usar PCA para factorizar texto era común antes de los años 2000.
- Las versiones de este algoritmo se denominaban bajo la etiqueta de análisis semántico latente.
- Sin embargo, esto cambió con la introducción del modelado de tópicos, también conocido como Asignación Latente de Dirichlet (LDA), por Blei et al. en 2003.
- Estos autores propusieron tomar en serio la representación de bolsa de palabras y modelar los conteos de tokens como realizaciones de un marco probabilístico.

Topic Models

El Problema



Topic Models

El Problema

doc1
galaxy
planet
ball
ball
ball

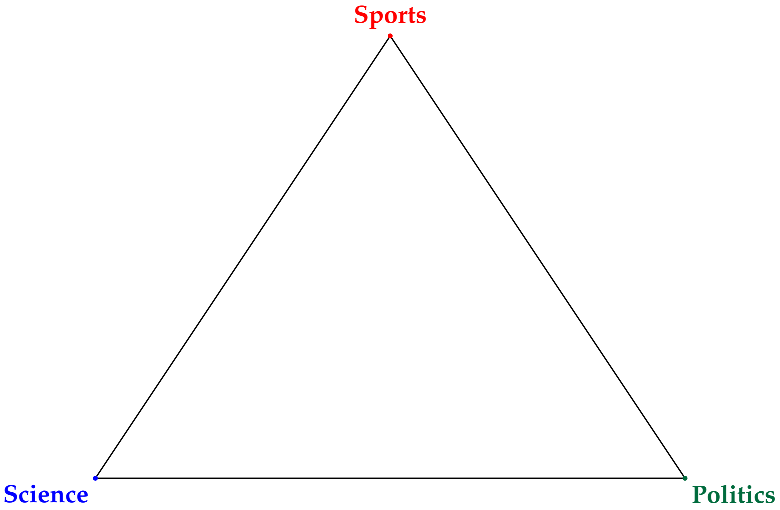
doc2
referendum
referendum
planet
planet
referendum

doc3
ball
planet
galaxy
planet
planet

doc4
ball
planet
referendum
galaxy
planet

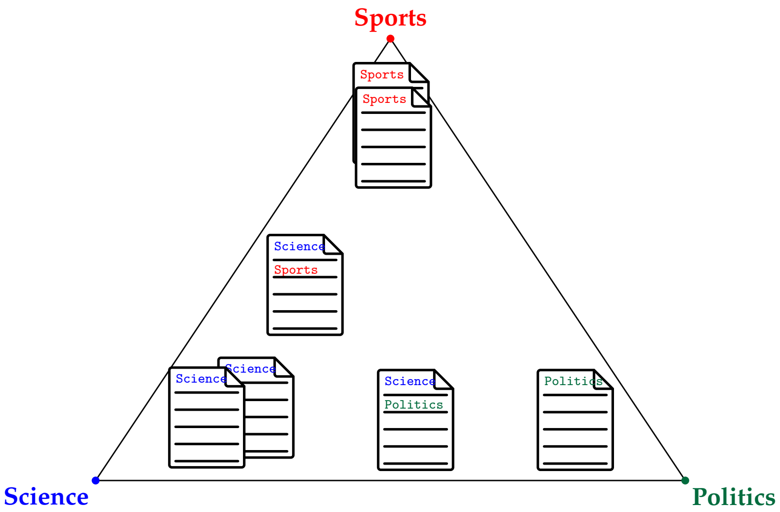
Topic Models

Latent Dirichlet Allocation



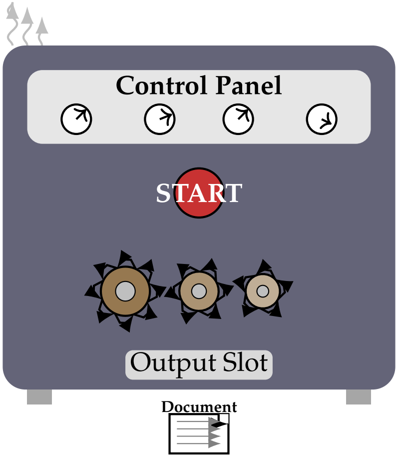
Topic Models

Latent Dirichlet Allocation



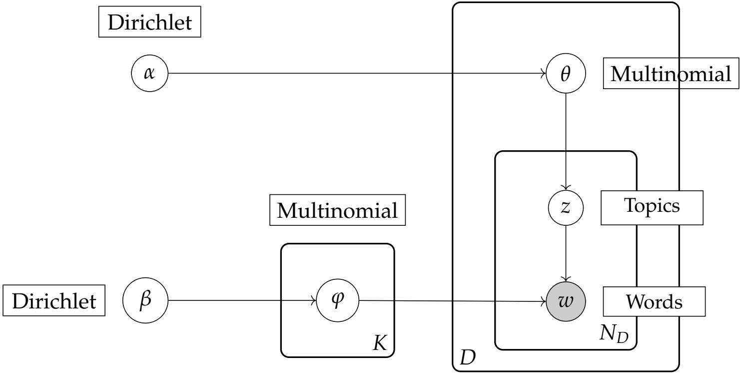
Topic Models

Latent Dirichlet Allocation



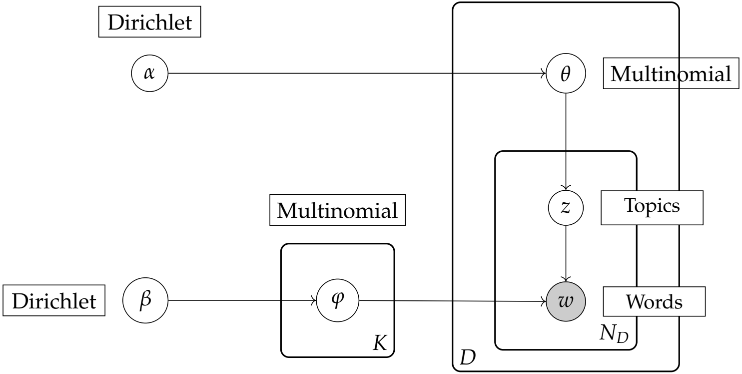
Topic Models

Latent Dirichlet Allocation: Blueprint



Topic Models

Latent Dirichlet Allocation: Blueprint



$$P(W, Z, \theta, \varphi; \alpha, \beta) = \prod_{j=1}^{M} P(\theta_j; \alpha) \prod_{i=1}^{K} P(\varphi_i; \beta) \prod_{t=1}^{N} P(Z_{j,t} \mid \theta_j) P(W_{j,t} \mid \varphi Z_{j,t})$$

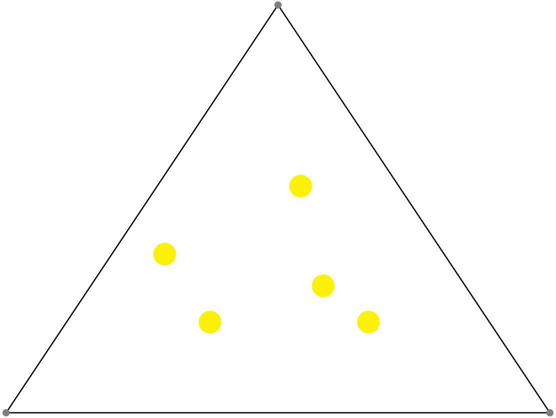
Topic Models

Latent Dirichlet Allocation: Blueprint

$$P(W, Z, \theta, \varphi; \alpha, \beta) = \prod_{j=1}^{M} P(\theta_j; \alpha) \prod_{i=1}^{K} P(\varphi_i; \beta) \prod_{t=1}^{N} P(Z_{j,t} \mid \theta_j) P(W_{j,t} \mid \varphi Z_{j,t})$$

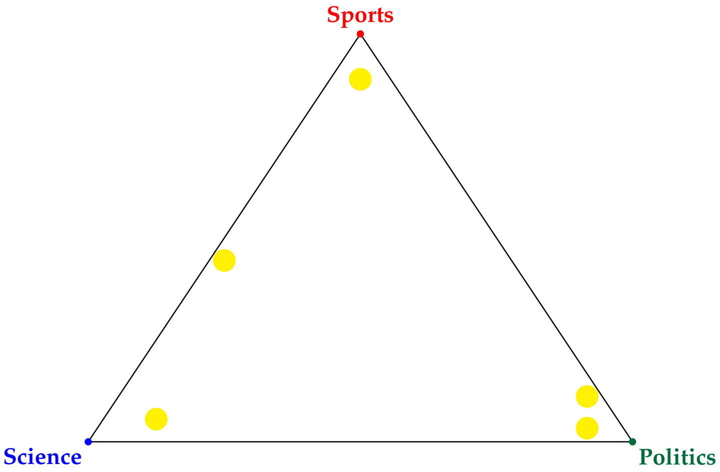
Topic Models

Dirichlet Distributions



Topic Models

Dirichlet Distributions



Topic Models

Dirichlet Distributions

- The Dirichlet distribution lives on something called the $(K-1)$**-simplex**. For our triangular house (when $K=3$):

$$\Delta^{K-1} = \left\{ \boldsymbol{\theta} \in \mathbb{R}^K : \theta_k \geq 0 \ \forall k, \ \sum_{k=1}^{K} \theta_k = 1 \right\}$$

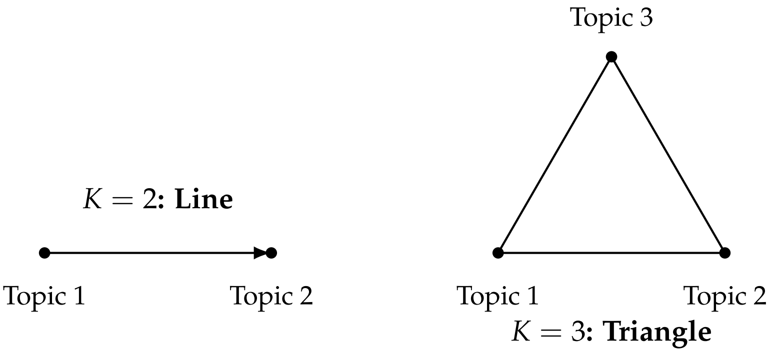
- This just says: $\boldsymbol{\theta}$ is a vector of K non-negative numbers that sum to 1
- We write $\boldsymbol{\theta} \sim \text{Dir}(\boldsymbol{\alpha})$ where $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)$ is the concentration vector.
- **The probability density function is:**

$$f(\boldsymbol{\theta} \mid \boldsymbol{\alpha}) = \frac{1}{B(\boldsymbol{\alpha})} \prod_{k=1}^{K} \theta_k^{\alpha_k - 1}$$

- where $B(\boldsymbol{\alpha}) = \frac{\prod_{k=1}^{K} \Gamma(\alpha_k)}{\Gamma(\alpha_0)}$ and $\alpha_0 = \sum_{k=1}^{K} \alpha_k$.
- $B(\boldsymbol{\alpha})$ is just a normalizing constant that ensures probabilities sum to 1.

Topic Models

Dirichlet Distributions



Topic Models

Dirichlet Distributions

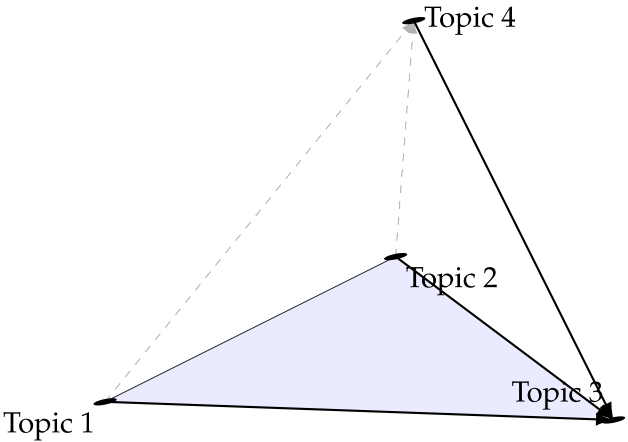


Figure 1: A tetrahedron for $K = 4$ topics. For $K > 4$, we enter higher dimensions!

Topic Models

Two Dirichlet Distributions in LDA

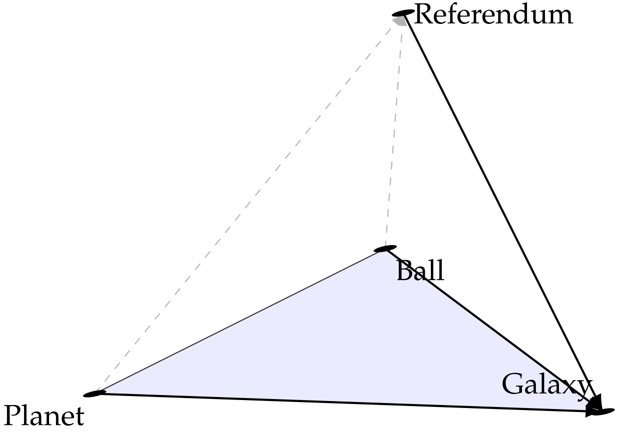
$$P(W, Z, \theta, \varphi; \alpha, \beta) = \prod_{j=1}^{M} P(\theta_j; \alpha) \prod_{i=1}^{K} P(\varphi_i; \beta) \prod_{t=1}^{N} P(Z_{j,t} \mid \theta_j) P(W_{j,t} \mid \varphi Z_{j,t})$$

LDA actually uses the Dirichlet distribution **twice**:

1. **Document-Topic distribution** ($\boldsymbol{\theta} \sim \text{Dir}(\boldsymbol{\alpha})$):
   - Lives on a triangle with corners = topics (Sports, Science, Politics)
   - Each point = a document's topic mixture
   - Usually sparse ($\alpha < 1$) because documents focus on one or two topics
2. **Topic-Word distribution** ($\boldsymbol{\varphi} \sim \text{Dir}(\boldsymbol{\beta})$):
   - Lives on a tetrahedron with corners = words (ball, planet, galaxy, referendum)
   - Each point = a topic's word distribution

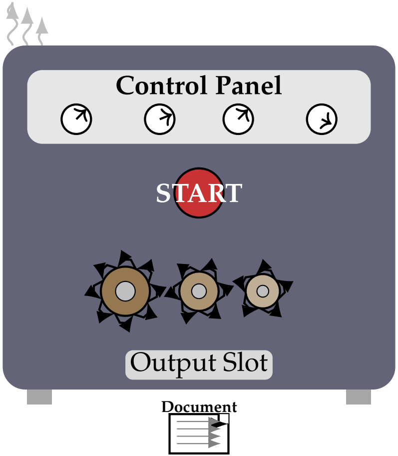
Topic Models

Two Dirichlet Distributions in LDA



Topic Models

LDA



Topic Models
LDA

$$P(W,Z,\theta,\varphi;\alpha,\beta)=\prod_{j=1}^{M}P(\theta_j;\alpha)\prod_{i=1}^{K}P(\varphi_i;\beta)\prod_{t=1}^{N}P(Z_{j,t}\mid\theta_j)P(W_{j,t}\mid\varphi Z_{j,t})$$

Topic Models: Examples LDA